

BLADE: Bilevel Low-rank Augmented-Lagrangian Driven Erasure for LLM Unlearning

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1 Method

We formulate LLM unlearning as constrained bilevel optimization. The inner loop preserves retain performance and the outer loop drives forgetting subject to an augmented Lagrangian (ALM) constraint. We confine all updates to LoRA adapters attached to a frozen base model.

1.1 Problem Setup

Let M be a pretrained LLM with parameters θ_0 , $\mathcal{D}_{\text{forget}}$ the data to unlearn, and $\mathcal{D}_{\text{retain}}$ a set of data the model should preserve. We seek parameters θ such that (i) the model’s output distribution on $\mathcal{D}_{\text{forget}}$ is high-entropy (memorized content is forgotten) and (ii) the cross-entropy loss on $\mathcal{D}_{\text{retain}}$ remains below a threshold ε .

1.2 LoRA Parameterization

Rather than modifying the full weight matrices, we attach LoRA adapters (Hu et al., 2022) to all projection matrices ($q/k/v/o_proj$, $gate/up/down_proj$):

$$W' = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times d} \quad (1)$$

where $r \ll d$ is the rank and α/r is a scaling factor. With $r = 16$ on a 7B model, only $\sim 0.08\%$ of parameters are trainable. This has two practical benefits for unlearning: (a) the low-rank subspace limits the expressivity of the update and prevents catastrophic weight drift, and (b) disabling adapters instantly recovers the exact original model, which simplifies debugging and rollback.

1.3 Bilevel Formulation

We decompose the unlearning problem into two nested optimization levels with a deliberate assignment: *retain in the inner loop, forget in the outer loop*.

Inner loop (retain preservation). For each outer step, run K SGD steps on the retain set:

$$\theta \leftarrow \theta - \eta_{\text{in}} \nabla_{\theta} \mathcal{L}_{\text{CE}}(M\theta; \mathcal{B}_{\text{ret}}) \quad (2)$$

where $\mathcal{B}_{\text{ret}} \sim \mathcal{D}_{\text{retain}}$ is a mini-batch and $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy loss. The inner loop uses a fast learning rate ($\eta_{\text{in}} = 2\text{e-}4$) to rapidly restore any retain damage caused by the previous outer step.

Outer loop (constrained forgetting). The outer objective combines the forget loss with an inequality constraint on retain performance enforced via an augmented Lagrangian:

$$\mathcal{L}_{\text{outer}} = \mathcal{L}_{\text{fgt}} + \lambda r + \frac{\rho}{2} [\max(0, r)]^2, \quad r = L_{\text{ret}} - \varepsilon \quad (3)$$

where $L_{\text{ret}} = \mathcal{L}_{\text{CE}}(M\theta; \mathcal{B}'_{\text{ret}})$ is evaluated on a *fresh* retain batch (separate from the inner loop), λ is the Lagrange multiplier, ρ is the quadratic penalty coefficient, and ε is the retain budget. The outer loop uses a slow learning rate ($\eta_{\text{out}} = 3\text{e-}5$), ensuring that each forgetting step is small enough for the inner loop to repair.

Repair-first principle. This assignment—retain inside, forget outside—inverts prior bilevel unlearning approaches (e.g., SCRUB (Kurmanji et al., 2023) puts forgetting in the inner loop and retain repair in the outer loop). Our design reflects a key insight: forgetting is the *hard* direction that must be carefully metered, while retain repair is a *fast* operation that converges in few steps. The speed asymmetry ($\eta_{\text{in}}/\eta_{\text{out}} \approx 7\text{--}10\times$) guarantees that the model enters each outer step from a state with good retain performance, preventing the catastrophic retain collapse that occurs when aggressive forgetting outpaces repair.

1.4 Clamped Entropy Loss

Entropy maximization as a forget objective is well-known (Yuan et al., 2025), but unclamped formulations have a fundamental problem: they apply nonzero gradients to every token at every step, even those whose output distribution is already near-uniform. This produces unnecessarily large outer perturbations that destabilize bilevel optimization. We propose a qualitatively different objective—*clamped entropy*—that defines when a token is “sufficiently forgotten” and stops optimizing it:

$$\mathcal{L}_{\text{forget}} = \frac{1}{T} \sum_{t=1}^T \max(0, \tau \cdot H_{\max} - H(p_t)) \quad (4)$$

where $p_t = \text{softmax}(z_t)$ is the output distribution at position t , $H(p_t) = -\sum_v p_t(v) \log p_t(v)$ is its Shannon entropy, $H_{\max} = \log V$ is the entropy of a uniform distribution over vocabulary V , and $\tau \in (0, 1)$ is the target fraction.

Vocabulary-invariant threshold. The normalization by $H_{\max} = \log V$ makes τ interpretable across models: $\tau = 0.7$ means “70% of maximum possible uncertainty,” whether the vocabulary has 32K tokens (Llama-2) or 128K tokens (Llama-3). Without this normalization, a fixed entropy threshold would need to be re-tuned for each tokenizer.

Hard deadzone. The $\max(0, \cdot)$ clamp is the critical distinction from prior entropy-based objectives. Once $H(p_t) \geq \tau \cdot H_{\max}$, the gradient contribution of token t is *exactly zero*—not approximately small, not asymptotically vanishing, but identically zero. Tokens “drop out” of the loss as they are forgotten, creating three properties critical for stable bilevel unlearning:

- **Bounded loss:** $\mathcal{L}_{\text{forget}} \in [0, \tau \cdot H_{\max}]$. The outer perturbation magnitude is bounded by construction (Proposition 1), so the inner loop’s K steps reliably repair retain damage regardless of training stage.
- **Self-stabilizing:** the number of “active” tokens (those with $H(p_t) < \tau \cdot H_{\max}$) decreases monotonically during successful training. The effective gradient norm shrinks automatically, naturally decelerating the outer loop without learning rate scheduling.
- **Reference-free:** requires only a single forward pass through the current model, unlike

DPO-style losses (NPO, SimNPO) that need a frozen reference model in memory.

In contrast, unclamped entropy maximization (Yuan et al., 2025) applies gradient pressure to *all* tokens at *all* steps—including those already at 99% of $H_{\max}$ —wasting optimization budget and creating gratuitous perturbations to shared representations. Gradient ascent (Jang et al., 2023) is even worse: its loss is unbounded ($-\log p \rightarrow \infty$ as $p \rightarrow 0$), so a single highly-memorized token can dominate the entire gradient (see Proposition 1c).

1.5 Augmented Lagrangian with Asymmetric Dual Update

The Lagrange multiplier λ controls how much weight retain preservation gets in the outer objective. Instead of tuning λ as a fixed hyperparameter, we learn it via a dual update rule.

A critical design choice is the *type* of constraint. Standard ALM for equality constraints uses a symmetric update and a penalty that is always active (Bertsekas, 1999). We instead enforce an *inequality* constraint $L_{\text{ret}} \leq \varepsilon$: the quadratic penalty $\frac{\rho}{2}[\max(0, r)]^2$ activates *only when violated* (Eq. 3). This means the model is never penalized for achieving retain loss *below* ε —it is free to improve on retain as long as it doesn’t exceed the budget. On top of this, we introduce an *asymmetric* dual update:

$$\lambda \leftarrow \begin{cases} \lambda + \rho \cdot r & \text{if } r > 0 \text{ (violation)} \\ \lambda + 0.1\rho \cdot r & \text{if } r \leq 0 \text{ (satisfaction)} \end{cases} \quad (5)$$

The $10\times$ slower decay means that once a retain violation drives λ up, the system permanently increases retain protection. Without this asymmetry, symmetric updates let λ drop back between spikes, causing repeated forget–retain oscillation cycles.

Auto-epsilon. Instead of hand-tuning an absolute ε , we set it as a fraction of the model’s initial retain loss. Before training, we run one inner loop and compute:

$$\varepsilon = \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CE}}^{(k)} \quad (6)$$

where $\mathcal{L}_{\text{CE}}^{(k)}$ is the retain loss at inner step k . This way, ε adapts automatically to the model and dataset.

Algorithm 1 BLADE: Bilevel Augmented Lagrangian Unlearning

Require: Model M with frozen weights θ_0 , $\mathcal{D}_{\text{forget}}$, $\mathcal{D}_{\text{retain}}$

Require: Hyperparams: K , η_{in} , η_{out} , ε_{mul} , ρ , λ_{init} , τ , T

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1: Attach LoRA adapters to  $M$ ;  $\theta \leftarrow$  LoRA
   params
2: inner_opt  $\leftarrow$  SGD( $\theta, \eta_{\text{in}}$ ); outer_opt  $\leftarrow$ 
   Adam( $\theta, \eta_{\text{out}}$ )
3:  $\lambda \leftarrow \lambda_{\text{init}}$ 
4: Run  $K$  inner steps, set  $\varepsilon \leftarrow \varepsilon_{\text{mul}} \cdot$ 
   mean(inner_losses)
5: for  $t = 1$  to  $T$  do
6:   Inner loop:  $K$  SGD steps on  $\mathcal{D}_{\text{retain}}$ 
   (Eq. (2))
7:   Outer step:  $\nabla_{\theta} \mathcal{L}_{\text{outer}}$  via Eq. (3) with
    $\mathcal{L}_{\text{forget}}$  from Eq. (4)
8:    $\theta \leftarrow \theta - \eta_{\text{out}} \cdot \text{Adam}(\nabla_{\theta} \mathcal{L}_{\text{outer}})$ 
9:   Dual update:  $\lambda$  via Eq. (5)
10: end for
11: return  $M$  with trained LoRA adapters
```

1.6 Algorithm Summary

2 Results

2.1 Benchmarks and Metrics

We evaluate on four benchmarks: **TOFU** (Maini et al., 2024) (fictitious QA, Llama-3.2-1B/3B, 3 forget splits), **MUSE Books/News** (Shi et al., 2025) (Harry Potter / BBC articles, Llama-2-7B), **KnowUndo** (Tian et al., 2024) (copyright/privacy, Llama-2-7B-chat). All results are 5-fold (seeds 42/123/456/789/1024) unless noted.

Composite scores:

- TOFU: HM = hmean(MU, 1-Prob, 1-ROUGE)
- MUSE: HM = hmean(1-fk, 1-vm, rk)
- KnowUndo: HM = hmean(1-fgt_R, ret_R, MMLU)
- LLM Judge: HM_J = hmean(1-FL/2, RA/2, rRQ/2)

LLM judge (Claude Opus 4.7) scores on 0–2 scale: Forget Leakage (FL↓), Retain Accuracy (RA↑), retain Response Quality (rRQ↑).

2.2 TOFU (Llama-3.2-1B-Instruct)

2.3 TOFU (Llama-3.2-3B-Instruct)

2.4 MUSE Books (Llama-2-7B)

2.5 MUSE News (Llama-2-7B)

2.6 KnowUndo (Llama-2-7B-chat)

All methods 5-fold except BLURNPO (OOM). MMLU evaluated once at seed=42, used as constant for HM computation.

2.7 Sustainability and Scalability (MUSE News)

Scalability: single-shot unlearning on progressively larger forget sets (1–4× the 889-sample standard). Sustainability: sequential unlearning where each step applies the method on a new 889-sample forget set starting from the previous checkpoint.

PDU collapses at 4× scale (HM=0.005, retain→0.002) and after 2 sequential steps (HM drops from 0.58 to 0.02). BLADE maintains stable performance: scalability HM stays 0.542–0.552 through 3×, and sustainability HM holds 0.542–0.525 through 4 sequential steps.

2.8 Ablation Study

Key findings: (1) ALM is essential—removing it collapses the model (A16, HM=0.233). (2) LoRA provides critical implicit regularization (A15, ret drops 0.658→0.306). (3) Clamped entropy is the only viable forget loss; GA (A4), NPO (A5), and logit margin (A17) all fail catastrophically even with full bilevel ALM. (4) Inner loop adds safety margin (+0.027 retain) but ALM alone works reasonably (A1). (5) Bilevel assignment matters: swapping (retain outer, forget inner) causes λ explosion (A18, vm=0.643).

2.9 Training Dynamics

BLADE follows a consistent three-phase pattern across all settings: (1) **Warm-up**: forget loss declines while retain stays near ε . (2) **Spike and ratchet**: accumulated forgetting disrupts shared representations; λ jumps from ~ 1 to ~ 2.2 – 2.6 , permanently increasing retain protection. (3) **Smooth convergence**: forget continues to zero, no further retain violations.

PDU’s dynamics contrast sharply: unbounded forget loss (starts at 700+), recurring

Table 1: TOFU 1B full results (5 seeds). HM = hmean(MU, 1-Prob, 1-ROUGE). Best in **bold**.

Method	Forget 1%				Forget 5%				Forget 10%			
	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑
Gold	.597	.166	.414	.655	.599	.127	.383	.676	.591	.116	.379	.677
GradAscent	.596±.00	.477±.01	.456±.01	.553±.01	.008±.01	.003±.01	.112±.08	.022±.04	.000±.00	.000±.00	.001±.00	.000±.00
GradDiff	.581±.01	.421±.04	.464±.02	.564±.01	.453±.00	.073±.00	.375±.01	.614±.01	.435±.00	.047±.00	.339±.01	.617±.00
NPO	.596±.00	.474±.01	.438±.02	.560±.01	.454±.01	.245±.01	.308±.01	.603±.01	.393±.03	.213±.01	.210±.01	.590±.02
SimNPO	.594±.00	.858±.01	.734±.02	.240±.01	.596±.00	.848±.00	.741±.01	.248±.00	.597±.00	.842±.00	.734±.01	.255±.00
RMU	.557±.00	.414±.01	.415±.00	.576±.00	.546±.00	.369±.01	.423±.01	.583±.00	.572±.00	.106±.02	.322±.01	.691±.01
BLURNPO	.598±.00	.676±.01	.588±.02	.417±.01	.518±.02	.475±.07	.407±.05	.541±.04	.089±.14	.087±.04	.253±.09	.154±.20
PDU	.602±.00	.186±.01	.313±.01	.690±.01	.588±.00	.073±.02	.214±.03	.740±.01	.592±.00	.004±.00	.065±.01	.797±.00
BLADE	.599±.00	.002±.00	.038±.00	.808±.00	.597±.00	.015±.00	.054±.01	.800±.00	.593±.01	.009±.00	.039±.01	.803±.00

Table 2: LLM Judge on TOFU 1B (5 seeds, Claude Opus 4.7). Best in **bold**.

Method	Forget 1%				Forget 5%				Forget 10%			
	FL↓	RA↑	rRQ↑	HM _J ↑	FL↓	RA↑	rRQ↑	HM _J ↑	FL↓	RA↑	rRQ↑	HM _J ↑
GradAscent	1.11±.05	1.66±.01	1.99±.00	.674±.02	0.07±.07	0.10±.09	0.51±.43	.117±.10	0.00±.00	0.00±.00	0.00±.00	.000±.00
GradDiff	1.21±.07	1.54±.04	1.97±.01	.620±.02	0.65±.03	1.01±.01	1.68±.01	.643±.00	0.46±.01	0.91±.02	1.53±.04	.625±.01
NPO	1.05±.08	1.67±.01	1.99±.00	.696±.03	0.44±.02	1.14±.05	1.89±.02	.731±.01	0.44±.02	0.94±.07	1.82±.04	.665±.03
SimNPO	1.76±.03	1.66±.01	1.99±.00	.288±.03	1.58±.01	1.69±.01	1.99±.00	.435±.01	1.58±.02	1.72±.02	1.99±.00	.435±.02
RMU	0.81±.05	1.39±.01	1.93±.01	.721±.01	0.72±.04	1.36±.02	1.91±.01	.736±.01	0.29±.04	1.50±.01	1.98±.00	.854±.01
BLURNPO	1.40±.05	1.67±.01	1.99±.00	.541±.03	0.86±.08	1.37±.06	1.98±.01	.709±.01	0.26±.24	0.36±.38	0.73±.64	.255±.21
PDU	0.86±.06	1.60±.01	1.97±.00	.746±.02	0.37±.08	1.56±.02	1.96±.01	.850±.01	0.04±.01	1.57±.01	1.98±.01	.906±.00
BLADE	0.02±.02	1.65±.01	1.98±.01	.929±.01	0.07±.02	1.64±.02	1.98±.01	.919±.00	0.04±.02	1.63±.02	1.98±.00	.919±.01

retain spikes throughout training, and no self-correcting mechanism.

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Table 3: TOFU 3B full results (5 seeds). Best in **bold**.

Method	Forget 1%				Forget 5%				Forget 10%			
	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑
Gold	.663	.179	.409	.656	.659	.130	.388	.698	.661	.115	.382	.698
GradAscent	.668 $\pm$.00	.564 $\pm$.01	.525 $\pm$.01	.509 $\pm$.01	.481 $\pm$.02	.140 $\pm$.02	.333 $\pm$.01	.632 $\pm$.01	.000 $\pm$.00	.000 $\pm$.00	.003 $\pm$.00	.000 $\pm$.00
GradDiff	.659 $\pm$.00	.569 $\pm$.02	.578 $\pm$.03	.483 $\pm$.02	.566 $\pm$.01	.147 $\pm$.01	.395 $\pm$.01	.653 $\pm$.00	.532 $\pm$.02	.079 $\pm$.01	.361 $\pm$.02	.662 $\pm$.01
NPO	.668 $\pm$.00	.553 $\pm$.01	.494 $\pm$.01	.525 $\pm$.01	.542 $\pm$.01	.219 $\pm$.00	.359 $\pm$.01	.640 $\pm$.00	.549 $\pm$.01	.237 $\pm$.01	.353 $\pm$.06	.640 $\pm$.02
SimNPO	.656 $\pm$.00	.920 $\pm$.01	.830 $\pm$.02	.151 $\pm$.01	.657 $\pm$.00	.901 $\pm$.00	.818 $\pm$.01	.175 $\pm$.00	.652 $\pm$.00	.889 $\pm$.00	.808 $\pm$.00	.191 $\pm$.00
RMU	.657 $\pm$.00	.861 $\pm$.00	.715 $\pm$.01	.246 $\pm$.00	.644 $\pm$.00	.602 $\pm$.00	.534 $\pm$.00	.483 $\pm$.00	.647 $\pm$.00	.390 $\pm$.01	.472 $\pm$.00	.591 $\pm$.00
BLURNPO	.667 $\pm$.00	.822 $\pm$.02	.788 $\pm$.02	.253 $\pm$.02	.640 $\pm$.02	.683 $\pm$.08	.593 $\pm$.09	.412 $\pm$.08	.385 $\pm$.17	.245 $\pm$.16	.314 $\pm$.09	.502 $\pm$.10
PDU	.692 $\pm$.00	.168 $\pm$.01	.283 $\pm$.01	.742 $\pm$.00	.687 $\pm$.00	.009 $\pm$.00	.085 $\pm$.01	.843 $\pm$.00	.680 $\pm$.01	.000 $\pm$.00	.029 $\pm$.01	.857 $\pm$.00
BLADE	.654 $\pm$.00	.000 $\pm$.00	.015 $\pm$.01	.846 $\pm$.00	.655 $\pm$.00	.007 $\pm$.01	.028 $\pm$.02	.842 $\pm$.01	.647 $\pm$.00	.005 $\pm$.00	.038 $\pm$.01	.836 $\pm$.01

Table 4: LLM Judge on TOFU 3B (5 seeds, Claude Opus 4.7). Best in **bold**.

Method	Forget 1%				Forget 5%				Forget 10%			
	FL↓	RA↑	rRQ↑	HM _J ↑	FL↓	RA↑	rRQ↑	HM _J ↑	FL↓	RA↑	rRQ↑	HM _J ↑
GradAscent	1.34 $\pm$.09	1.85 $\pm$.00	2.00 $\pm$.00	.587 $\pm$.05	0.69 $\pm$.04	1.45 $\pm$.02	1.98 $\pm$.01	.766 $\pm$.01	0.00 $\pm$.00	0.00 $\pm$.00	0.00 $\pm$.00	.000 $\pm$.00
GradDiff	1.50 $\pm$.07	1.82 $\pm$.02	2.00 $\pm$.00	.490 $\pm$.04	0.83 $\pm$.01	1.29 $\pm$.02	1.71 $\pm$.06	.677 $\pm$.01	0.66 $\pm$.03	1.16 $\pm$.06	1.43 $\pm$.20	.648 $\pm$.04
NPO	1.23 $\pm$.04	1.84 $\pm$.00	2.00 $\pm$.00	.640 $\pm$.02	0.74 $\pm$.04	1.55 $\pm$.01	2.00 $\pm$.00	.774 $\pm$.01	0.61 $\pm$.08	1.52 $\pm$.04	1.99 $\pm$.01	.796 $\pm$.01
SimNPO	1.83 $\pm$.05	1.84 $\pm$.01	2.00 $\pm$.00	.220 $\pm$.06	1.74 $\pm$.01	1.85 $\pm$.01	2.00 $\pm$.00	.305 $\pm$.01	1.71 $\pm$.01	1.86 $\pm$.01	2.00 $\pm$.00	.331 $\pm$.01
RMU	1.59 $\pm$.04	1.81 $\pm$.00	2.00 $\pm$.00	.432 $\pm$.03	1.11 $\pm$.03	1.70 $\pm$.01	1.99 $\pm$.00	.679 $\pm$.01	0.85 $\pm$.02	1.70 $\pm$.01	1.99 $\pm$.00	.765 $\pm$.01
BLURNPO	1.73 $\pm$.12	1.85 $\pm$.00	2.00 $\pm$.00	.308 $\pm$.11	1.40 $\pm$.15	1.71 $\pm$.05	2.00 $\pm$.00	.538 $\pm$.08	0.73 $\pm$.35	1.05 $\pm$.51	1.51 $\pm$.60	.550 $\pm$.16
PDU	0.62 $\pm$.04	1.72 $\pm$.01	1.97 $\pm$.00	.828 $\pm$.01	0.07 $\pm$.01	1.75 $\pm$.01	1.98 $\pm$.00	.941 $\pm$.00	0.01 $\pm$.01	1.73 $\pm$.03	1.95 $\pm$.04	.940 $\pm$.01
BLADE	0.01 $\pm$.01	1.85 $\pm$.00	2.00 $\pm$.00	.970 $\pm$.00	0.02 $\pm$.03	1.82 $\pm$.01	1.99 $\pm$.00	.962 $\pm$.00	0.03 $\pm$.03	1.84 $\pm$.01	1.99 $\pm$.00	.965 $\pm$.00

Table 5: MUSE Books (5-fold except BLADE/BLURNPO seed=42). HM = hmean(1−fk, 1−vm, rk). Best in **bold** (excl. Gold).

Method	fk↓	vm↓	rk↑	HM↑
Gold (retrain)	.303	.145	.687	.739
GradAscent	.000 $\pm$.00	.000 $\pm$.00	.000 $\pm$.00	.000 $\pm$.00
GradDiff	.000 $\pm$.00	.000 $\pm$.00	.004 $\pm$.00	.011 $\pm$.01
NPO	.303 $\pm$.02	.342 $\pm$.03	.574 $\pm$.02	.638 $\pm$.01
SimNPO	.238 $\pm$.02	.002 $\pm$.00	.600 $\pm$.01	.753 $\pm$.01
RMU	.210 $\pm$.00	.113 $\pm$.01	.598 $\pm$.01	.738 $\pm$.00
PDU	.139 $\pm$.01	.129 $\pm$.01	.372 $\pm$.01	.600 $\pm$.01
BLURNPO [†]	.175	.000	.527	.730
BLADE [†]	.110	.000	.658	.823

[†]seed=42 only (BLURNPO OOM on 4×40GB for 5-fold).

Table 6: MUSE Books LLM judge (5-fold; [†]seed=42). Best in **bold**.

Method	FL↓	RA↑	rRQ↑	HM _J ↑
GradAscent	0.00 $\pm$.00	0.00 $\pm$.00	0.00 $\pm$.00	.000 $\pm$.00
GradDiff	0.00 $\pm$.00	0.04 $\pm$.02	0.01 $\pm$.01	.010 $\pm$.01
NPO	0.91 $\pm$.03	1.28 $\pm$.03	1.61 $\pm$.01	.647 $\pm$.01
SimNPO	0.31 $\pm$.12	1.28 $\pm$.02	1.62 $\pm$.01	.753 $\pm$.02
RMU	0.32 $\pm$.02	1.41 $\pm$.02	1.68 $\pm$.01	.791 $\pm$.00
PDU	0.53 $\pm$.02	0.99 $\pm$.02	1.15 $\pm$.02	.586 $\pm$.01
BLURNPO [†]	0.21	1.09	1.47	.696
BLADE [†]	0.14	1.40	1.69	.814

Table 7: MUSE News (5-fold; [†]seed=42). Best HM in **bold**.

Method	fk↓	vm↓	rk↑	HM↑
Gold (retrain)	.324	.204	.552	.660
GradDiff	.329 $\pm$.03	.043 $\pm$.02	.269 $\pm$.02	.479 $\pm$.02
NPO	.517 $\pm$.02	.274 $\pm$.02	.435 $\pm$.01	.522 $\pm$.01
SimNPO	.628 $\pm$.01	.542 $\pm$.01	.513 $\pm$.01	.440 $\pm$.00
RMU	.495 $\pm$.01	.267 $\pm$.01	.434 $\pm$.01	.531 $\pm$.00
PDU	.525 $\pm$.02	.092 $\pm$.07	.508 $\pm$.04	.577 $\pm$.01
BLURNPO [†]	.317	.181	.304	.448
BLADE [†]	.550	.173	.475	.542

Table 8: MUSE News LLM judge (5-fold; [†]seed=42). Best in **bold**.

Method	FL↓	RA↑	rRQ↑	HM _J ↑
GradDiff	0.58 $\pm$.02	0.79 $\pm$.04	1.16 $\pm$.14	.529 $\pm$.03
NPO	1.20 $\pm$.03	1.05 $\pm$.02	1.58 $\pm$.01	.531 $\pm$.01
SimNPO	1.49 $\pm$.01	1.24 $\pm$.01	1.72 $\pm$.01	.447 $\pm$.01
RMU	1.13 $\pm$.01	1.15 $\pm$.02	1.60 $\pm$.01	.565 $\pm$.00
PDU	0.87 $\pm$.12	1.18 $\pm$.05	1.64 $\pm$.05	.638 $\pm$.02
BLURNPO [†]	0.47	0.51	1.47	.455
BLADE [†]	0.85	0.88	1.72	.579

Table 9: KnowUndo Copyright (5-fold; $\dagger$ seed=42). HM = hmean(1-fgt_R, ret_R, MMLU). Best in **bold**.

Method	Automated Metrics					LLM Judge (Haiku 3.5)			
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	HM↑	FL↓	RA↑	rRQ↑	HM _J ↑
GradAscent	.002±.00	.000±.00	.002±.00	.000±.00	—	0.00±.00	0.00±.00	0.00±.00	—
GradDiff	.002±.00	.000±.00	.642±.04	.199±.01	.348±.01	0.06±.07	0.96±.16	0.84±.14	.47±.06
NPO	.678±.02	.189±.01	.763±.02	.289±.00	.433±.00	1.37±.45	1.77±.10	1.86±.16	.49±.15
SimNPO	.303±.05	.048±.04	.788±.01	.317±.00	.461±.01	0.29±.22	1.76±.13	1.85±.18	.80±.04
RMU	.435±.01	.076±.00	.763±.00	.201±.01	.347±.01	0.42±.22	1.12±.12	1.42±.10	.62±.03
PDU	.045±.00	.011±.00	.836±.00	.285±.01	.442±.01	0.08±.04	1.61±.07	1.58±.13	.75±.03
BLURNPO $\dagger$	.494	.023	.689	.140	.287	0.01	0.24	0.32	—
BLADE	.189 ±.14	.014 ±.01	.840 ±.01	.340 ±.00	.482 ±.00	0.24±.19	1.81 ±.03	1.89 ±.13	.82 ±.03

Table 10: KnowUndo Privacy (5-fold; $\dagger$ seed=42). HM = hmean(1-fgt_R, ret_R, MMLU). Best in **bold**.

Method	Automated Metrics					LLM Judge (Haiku 3.5)			
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	HM↑	FL↓	RA↑	rRQ↑	HM _J ↑
GradAscent	.017±.02	.004±.01	.012±.01	.006±.01	—	0.00±.00	0.00±.00	0.00±.00	—
GradDiff	.049±.02	.039±.02	.708±.03	.406±.02	.516±.01	0.04±.02	1.08±.07	1.28±.15	.50±.04
NPO	.646±.02	.322±.02	.790±.00	.426±.01	.496±.00	0.99±.23	1.40±.14	1.97±.02	.66±.04
SimNPO	.516±.04	.302±.03	.889±.01	.557±.01	.546±.01	0.79±.23	1.55±.12	1.89±.04	.66±.02
RMU	.568±.01	.072±.00	.661±.01	.079±.01	.182±.01	0.13±.06	0.17±.07	0.52±.15	.18±.07
PDU	.055±.02	.022±.01	.813±.02	.452±.02	.545±.01	0.05±.03	1.25±.06	1.53±.04	.58±.02
BLURNPO $\dagger$	.103	.038	.704	.362	.498	0.04	0.42	0.84	—
BLADE	.194 ±.03	.075 ±.03	.930 ±.02	.623 ±.03	.602 ±.01	0.20±.12	1.50 ±.07	1.74 ±.09	.66 ±.03

Table 11: Scalability on MUSE News.

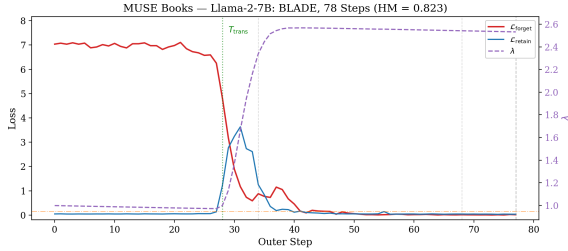
Scale	Method	fk↓	vm↓	rk↑	HM↑
1× (889)	PDU	.542	.065	.522	.581
	BLADE	.550	.173	.475	.542
2× (1778)	PDU	.495	.050	.467	.579
	BLADE	.504	.321	.500	.547
3× (2667)	PDU	.538	.024	.488	.573
	BLADE	.466	.366	.505	.552
4× (3554)	PDU	.321	.021	.002	.005
	BLADE		(running)		

Table 12: Sustainability on MUSE News (sequential unlearning steps).

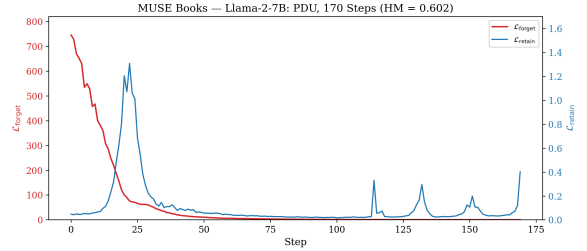
Step	Method	fk↓	vm↓	rk↑	HM↑
1	PDU	.542	.065	.522	.580
	BLADE	.550	.173	.475	.542
2	PDU	.554	.007	.470	.558
	BLADE	.539	.212	.485	.545
3	PDU	.550	.001	.005	.016
	BLADE	.556	.230	.459	.523
4	PDU	.591	.003	.051	.130
	BLADE	.564	.222	.469	.525

Table 13: Ablation on MUSE Books (seed=42). Each row changes one component from the full BLADE. LLM judge scores on 0–2 scale.

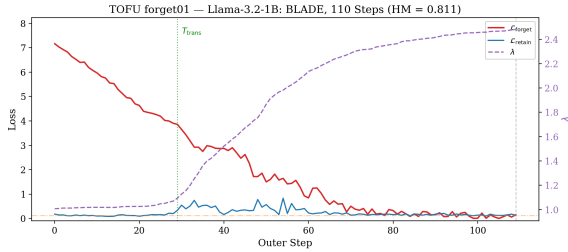
Configuration	fk↓	vm↓	ret↑	HM↑	Δret	FLJ↓	RAJ↑	rRQJ↑	HMJ↑
Full BLADE	.110	.000	.658	.823	—	0.14	1.40	1.69	.814
$K=0$ (no inner loop)	.072	.000	.631	.819	−.027	0.10	1.37	1.68	.811
$\tau=1.0$ (unclamped entropy)	.161	.003	.604	.780	−.054	0.20	1.34	1.64	.785
No LoRA (full fine-tuning)	.002	.000	.306	.459	−.352	0.00	0.70	1.25	.550
ALM off ($\lambda=0, \rho=0$)	.000	.002	.092	.233	−.566	0.01	0.24	0.28	.117
Swapped bilevel (forget inner)	.372	.643	.651	.506	−.007	—	—	—	—
GA forget loss	.006	.003	.060	.160	−.598	0.01	0.10	0.18	.093
NPO forget loss	.457	.997	.662	.009	+.004	1.44	1.47	1.72	.495
Logit margin forget loss	.000	.000	.000	.000	−.658	0.00	0.00	0.00	.000



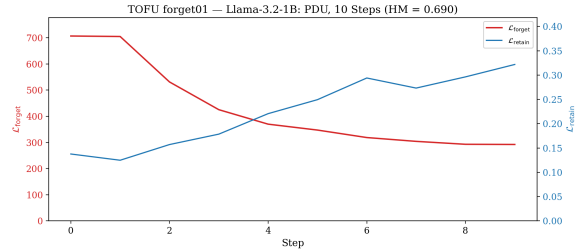
(a) BLADE on MUSE Books (HM = 0.823)



(b) PDU on MUSE Books (HM = 0.602)



(c) BLADE on TOFU 1B forget01 (HM = 0.811)



(d) PDU on TOFU 1B forget01 (HM = 0.690)

Figure 1: Training dynamics: BLADE (left) vs. PDU (right). BLADE shows three phases: warm-up, spike-and-ratchet, smooth convergence. PDU shows recurring retain oscillations with no correction.